

Question	Answer	Mark	Mark scheme	Additional guidance
18	52.5	P1	for start of process to find length of AC, eg $\frac{1}{2} \times 17 \times AC \times \sin 35 = 54$ or $(AC =) 54 \div (\frac{1}{2} \times 17 \times \sin 35) (= 11.076\dots)$	Check diagram for working throughout
		P1	for start of process to find AD or CD, eg $\frac{AD}{\sin 48} = \frac{"11.076"}{\sin 57}$ oe or $\frac{AD}{\sin 48} = \frac{[AC]}{\sin 57}$ oe or $\frac{CD}{\sin "75"} = \frac{"11.076"}{\sin 57}$ oe or $\frac{CD}{\sin "75"} = \frac{[AC]}{\sin 57}$ oe	Throughout "75" = 180 – 48 – 57 [AC] must be a numerical value and clearly identified by labelling or on the diagram with no contradiction.
		P1	for complete process to find AD or CD $(AD =) \frac{"11.076"}{\sin 57} \times \sin 48 (= 9.81\dots)$ or $(AD =) \frac{[AC]}{\sin 57} \times \sin 48$ or $(CD =) \frac{"11.076"}{\sin 57} \times \sin "75" (= 12.7\dots)$ or $(CD =) \frac{[AC]}{\sin 57} \times \sin "75"$	
		P1	for process to find area of triangle ACD, eg $\frac{1}{2} \times "11.076" \times "12.7" \times \sin 48$ or $\frac{1}{2} \times [AC] \times [CD] \times \sin 48$ or $\frac{1}{2} \times "11.076" \times "9.81" \times \sin "75"$ or $\frac{1}{2} \times [AC] \times [AD] \times \sin "75"$ or $\frac{1}{2} \times "9.81" \times "12.7" \times \sin 57$ or $\frac{1}{2} \times [AD] \times [CD] \times \sin 57$	[AC], [AD], [CD] must be numerical values and clearly identified by labelling or on the diagram with no contradiction.
		A1	for answer in the range 52.4 to 52.52	If an answer is shown in the range in working and then incorrectly rounded award full marks